When the compressors don't arrive

Liebherr CNc 5203 fridge-freezer · key component: compressor · Red Sea crisis vs. a second source

Team 4: Sustainable Supply Chain Management (MB4) · TH Köln ·
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QUESTION

When our key supplier fails, what happens to fridges per week, cost per fridge and CO_2 per fridge – and what should we do about it?

ANSWER

The Red Sea detour around the Cape stops our line for 73 h a year and adds 3.09 € per fridge. **A second source in Slovakia, by truck, keeps output at 807 fridges a week for only 1.12 € more than normal and saves 4.4 kg CO_2e per fridge.**

1 Product, line and supply chain

PRODUCT
Liebherr CNc 5203 · 69.7 kg · list price 713.45 € net

KEY COMPONENT
hermetic piston compressor · 7.2 kg · 1 per fridge

PLANT & PLAN
Ochsenhausen (DE) · 810 fridges / week · 1 shift

China
Jiaxipera, Jiaxing
14.15 € / compressor

Ship via Suez + truck · 35 days at sea · lead time 36 working days

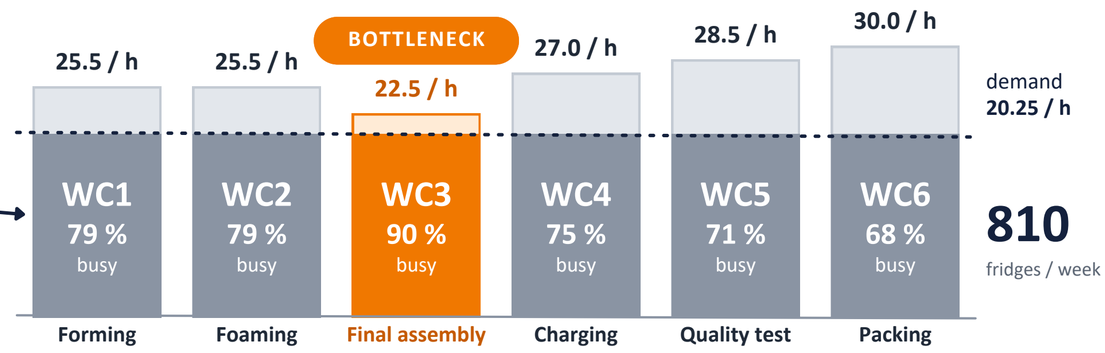
Crisis: around the Cape · 43 days at sea container freight +71 %

Slovakia
Embraco (Nidec), Spišská N. Ves
19.64 € / compressor (+39 %)

Truck · 983 km · 5.5 working days

Compressor stock
at the plant
order every Monday
safety stock for 95 % service level
1 compressor per fridge

Line capacity per hour vs. demand (dark part = busy share)



Why WC3? 5 places × 12 min × 90 % available = 22.5 fridges / h against a demand of 20.25 / h: busy 90 %, only 10 % spare. Random process times build the longest queue here in 47 of 52 weeks.

2 The disruption: Red Sea crisis 2024

Attacks on ships in the Red Sea forced container lines to sail around the Cape of Good Hope.

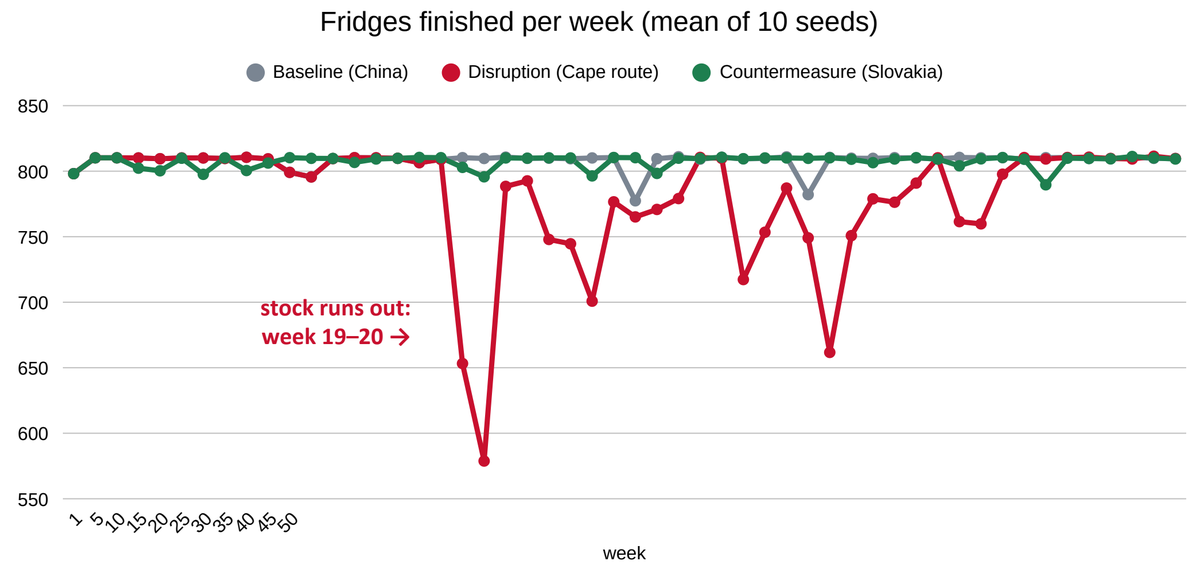
+8 days
at sea (35 → 43) and less reliable: sd 3.5 → 4.5 working days

+71 %
container Shanghai → Rotterdam: 3,091 € → 5,293 € (Drewry WCI)

–27
fridges per week on average over the year (808 → 781)

72.9 h
line stood still per year, instead of 4.4 h in normal times

How we modelled it: orders placed in weeks 10 to 35 take the Cape route, but planners keep ordering with the old 36-day lead time. The safety stock is used up and from week 19 the line starves.



3 Scoreboard

mean of seeds 1 to 10, [lowest to highest] below

Scenario	Fridges per week	Total cost per fridge	kg CO_2e per fridge	Hours line stood still
Baseline: China, via Suez normal operation	**808** 801 to 810	**292.88 €** 292.74 to 293.28	**21.3** 21.28 to 21.33	**4.4 h** 0 to 23
Disruption: Red Sea, Cape route orders of weeks 10 to 35	**781** 770 to 791	**295.97 €** 295.30 to 296.72	**21.7** 21.64 to 21.72	**72.9 h** 48 to 102
Countermeasure: Slovakia by truck disruption + second source	**807** 805 to 809	**294.00 €** 293.90 to 294.16	**16.9** 16.85 to 16.87	**6.2 h** 1 to 13
Alternative: +8 days stock (China) disruption + bigger buffer	**808** 804 to 809	**294.29 €** 294.10 to 294.50	**21.3** 21.28 to 21.33	**2.8 h** 0 to 11

Cost per fridge (baseline): materials 241.55 + station fixed costs 44.33 + transport 4.78 + holding 0.09 + CO_2 at 100 €/t 2.13 = 292.88 € (list price 713.45 € net). **Why stops in the baseline?** Safety stock covers 95 %, not 100 %, of the lead-time spread. Model: 52 weeks, one shift, 10 random seeds.

SOLUTION

Add Embraco Slovakia as a second source: compressors by truck in 5.5 instead of 36 working days

807
fridges / week in the crisis (781 without)

6.2 h
line stop per year (72.9 h without)

+1.12 €
per fridge vs normal (–1.97 € vs crisis)

–4.4 kg
CO_2e per fridge, –21 % vs China

Why it works: the truck lane avoids every sea chokepoint and its short, stable lead time needs far less safety stock. The CO_2 saving comes about half from the cleaner Slovak grid (95 vs 525 g/kWh) and half from shorter transport.

Is the insurance worth it?

- **Premium:** 1.12 € per fridge ≈ 47,000 € a year in normal times.
- **Risk:** one crisis year costs ≈ 1,390 fridges, up to 0.58 M € contribution margin at list price. It pays off if such a crisis hits about once in 12 years.
- **CO_2:** above ≈ 355 €/t (model: 100 €/t) Slovakia is cheaper even in normal times.
- **Bigger stock from China** stops the line less (2.8 h) but costs more (294.29 €), saves no CO_2 and fails if the detour lasts longer than 8 days.

4 Course concepts behind our results

each card: what the theory says → what our model shows

Bottleneck — see 1
Theory: the slowest station sets the line's output.
Our model: final assembly, 22.5 / h; longest queue in 47 of 52 weeks.

Variability & waiting — see 1
Theory: random times make queues grow near 100 % busy.
Our model: WC3 at 90 % busy already builds a queue.

Little's Law — check T3
Theory: WIP = throughput × lead time (L = λ × W).
Our model: 12.40 fridges = 20.1 / h × 0.62 h, consistent.

Lead time & safety stock — see 2
Theory: safety stock covers normal spread, not shocks.
Our model: +8 days at sea use it up: the line stops 73 h.

Total cost vs unit price — see 3
Theory: judge suppliers on total cost, not unit price.
Our model: compressor +39 %, total cost only +0.4 % (+1.12 €).

CO_2 per unit — see 3
Theory: most CO_2 sits in the purchased parts.
Our model: 16.5 of 21.3 kg in the compressor; Slovakia –4.4 kg.

5 Literature

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Scan for sources, data, papers & AI note